

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Triaged and resolved production incidents via root-cause analysis – traced **gRPC service deadlock** under concurrent load to shared resource acquisition conflicts; redesigned call sequencing for **100% evaluation success rate** at **sub-50ms p99**
- Maintained **distributed production systems** across gRPC, PostgreSQL, and ML serving layers – documented runbooks, participated in on-call rotation, applied post-mortem learnings to eliminate repeat failures
- Architected **vLLM inference engine with PagedAttention** – eliminated KV cache memory fragmentation under concurrent load; deployed with continuous batching achieving **zero OOM failures** at production traffic
- Built multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime, 44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Led backend, cloud infrastructure, and ML serving for a 3-person engineering team – designed, deployed, and operated distributed production systems end-to-end

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers for cross-process communication – selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99**
- Identified settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention – **96% latency reduction**, zero production regressions
- Built event-driven TypeScript/React system with observer pattern – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**; changes adopted into production branch
- Delivered design documents and code reviews at **95% test coverage**; reviewed by senior Chrome engineers and merged into stable channel

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Diagnosed **N+1 query bottleneck** degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records – profiled before fixing, not after
- Built stateless BaaS on AWS – JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users, 90% deployment time reduction** via CI/CD (GitHub Actions)

PROJECTS

Pulumi – Contributor | Go, TypeScript, IaC, CLI automation

github.com/pulumi

- Submitted Go CLI features and bug fixes for multi-cloud (AWS/Azure/GCP) provisioning – automation scripting at open-source scale; under review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability and correctness under concurrent operations and partial failures

SKILLS

Languages: Python, C++, TypeScript, JavaScript, Go, Rust, Java, SQL, Bash

Systems & Debugging: gRPC, Protocol Buffers, lock-free structures, root-cause analysis, on-call, runbooks, Linux, system administration

Testing & Automation: CI/CD (GitHub Actions), automation scripting, 95% test coverage, pytest, unit/integration testing

Monitoring & Observability: CloudWatch, Prometheus, Grafana, circuit breakers, health checks, auto-rollback

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, GCP

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor